OBJECTIVE OF THIS PPT.

- TO CONSTRUCT A LOGICAL APPROACH IN INVESTIGATING ANAEMIA
- TO UNDERSTAND THE PHYSIOLOGY OR PATHOLOGY BEHIND THE BASIC INVESTIGATIONS

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TABLE 59-1 Laboratory Tests in Anemia Diagnosis

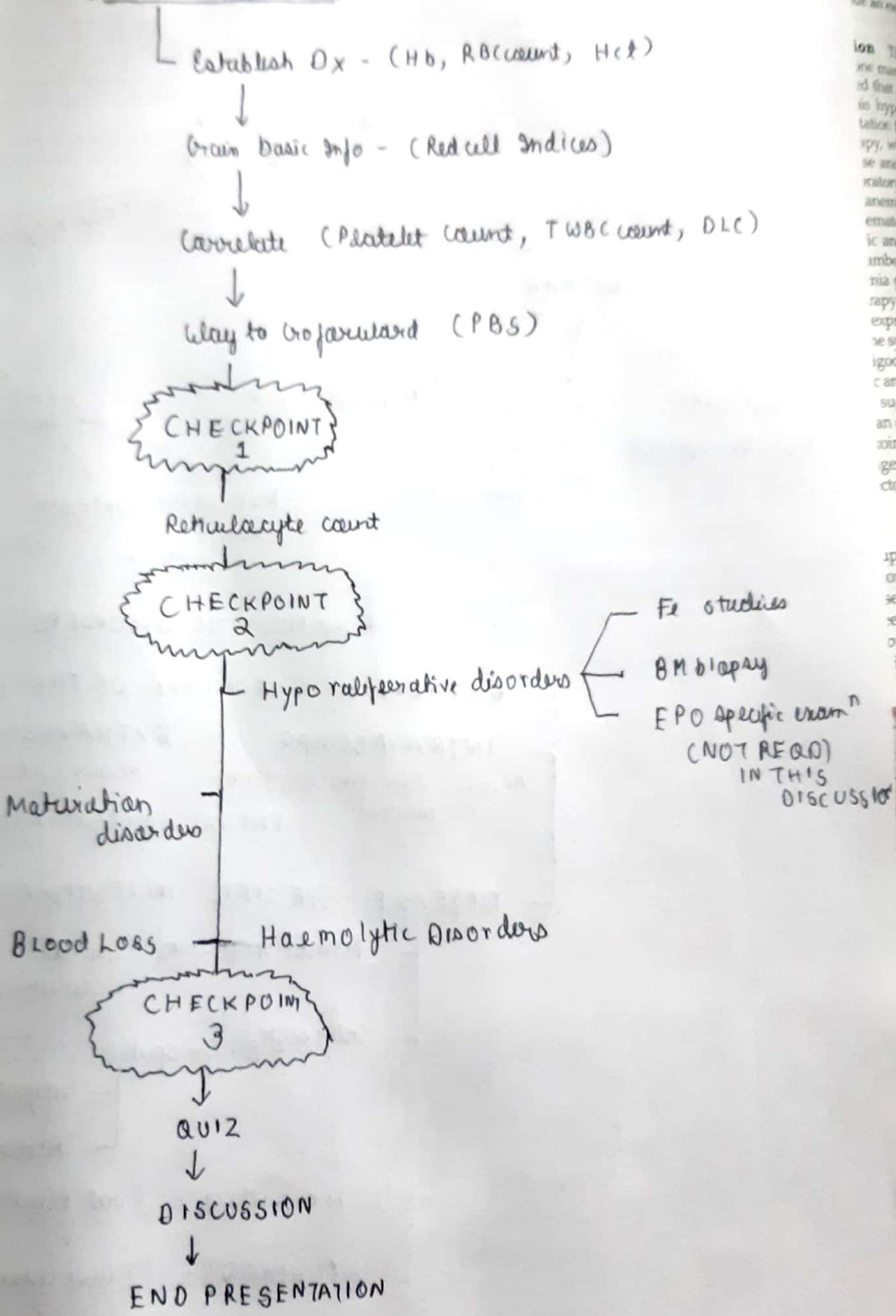
I. Complete blood count (CBC)
   A. Red blood cell count
      1. Hemoglobin
      2. Hematocrit
      3. Reticulocyte count
   B. Red blood cell indices
      1. Mean cell volume (MCV)
      2. Mean cell hemoglobin (MCH)
      3. Mean cell hemoglobin concentration (MCHC)
      4. Red cell distribution width (RDW)
   C. White blood cell count
      1. Cell differential
      2. Nuclear segmentation of neutrophils
   D. Platelet count
   E. Cell morphology
      1. Cell size
      2. Hemoglobin content
      3. Anisocytosis
      4. Poikilocytosis
      5. Polychromasia

II. Iron supply studies
   A. Serum iron
   B. Total iron-binding capacity
   C. Serum ferritin

II. Marrow examination
   A. Aspirate
      1. M/E ratio[a]
      2. Cell morphology
      3. Iron stain
   B. Biopsy
      1. Cellularity
      2. Morphology

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ANEMIA LAB INV.



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TABLE 59-3 Changes in Normal Hemoglobin/Hematocrit Values with Age, Sex, and Pregnancy

AGE/SEX	HEMOGLOBIN, g/dL	HEMATOCRIT, %
At birth	17	52
Childhood	12	36
Adolescence	13	40
Adult man	16 (±2)	47 (±6)
Adult woman (menstruating)	13 (±2)	40 (±6)
Adult woman (postmenopausal)	14 (±2)	42 (±6)
During pregnancy	12 (±2)	37 (±6)

Source: From RS Hillman et al: *Hematology in Clinical Practice*, 5th ed. New York, McGraw-Hill, 2010.

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8. *Count:*

At birth; 6-7 million/µL

Adults – Male: 5–6 million/µL

(average 5.5 million/µL)

Females: 4.5–5.5 million/µL

(average 4.8 million/µL)

Clinically 5 million/µL is taken as 100% RBC count.

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TABLE 59-2 Red Blood Cell Indices

INDEX	NORMAL VALUE
Mean cell volume (MCV) = (hematocrit × 10)/(red cell count × 10^6)	90 ± 8 fL
Mean cell hemoglobin (MCH) = (hemoglobin × 10)/(red cell count × 10^6)	30 ± 3 pg
Mean cell hemoglobin concentration = (hemoglobin × 10)/hematocrit, or MCH/MCV	33 ± 2%

RDW- $\left(\frac{S.D}{MCV} \times 100\right)$ 11–15%

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$$\text{MCV (fl)} = \frac{\text{PCV}}{\text{RBC}} \text{ X } 10$$

$$\text{MCH (pg)} = \frac{\text{Hb}}{\text{RBC}} \text{ X } 10$$

$$\text{MCHC (\%)} = \frac{\text{Hb (100mg blood)}}{\text{PCV}} \text{ X } 100$$

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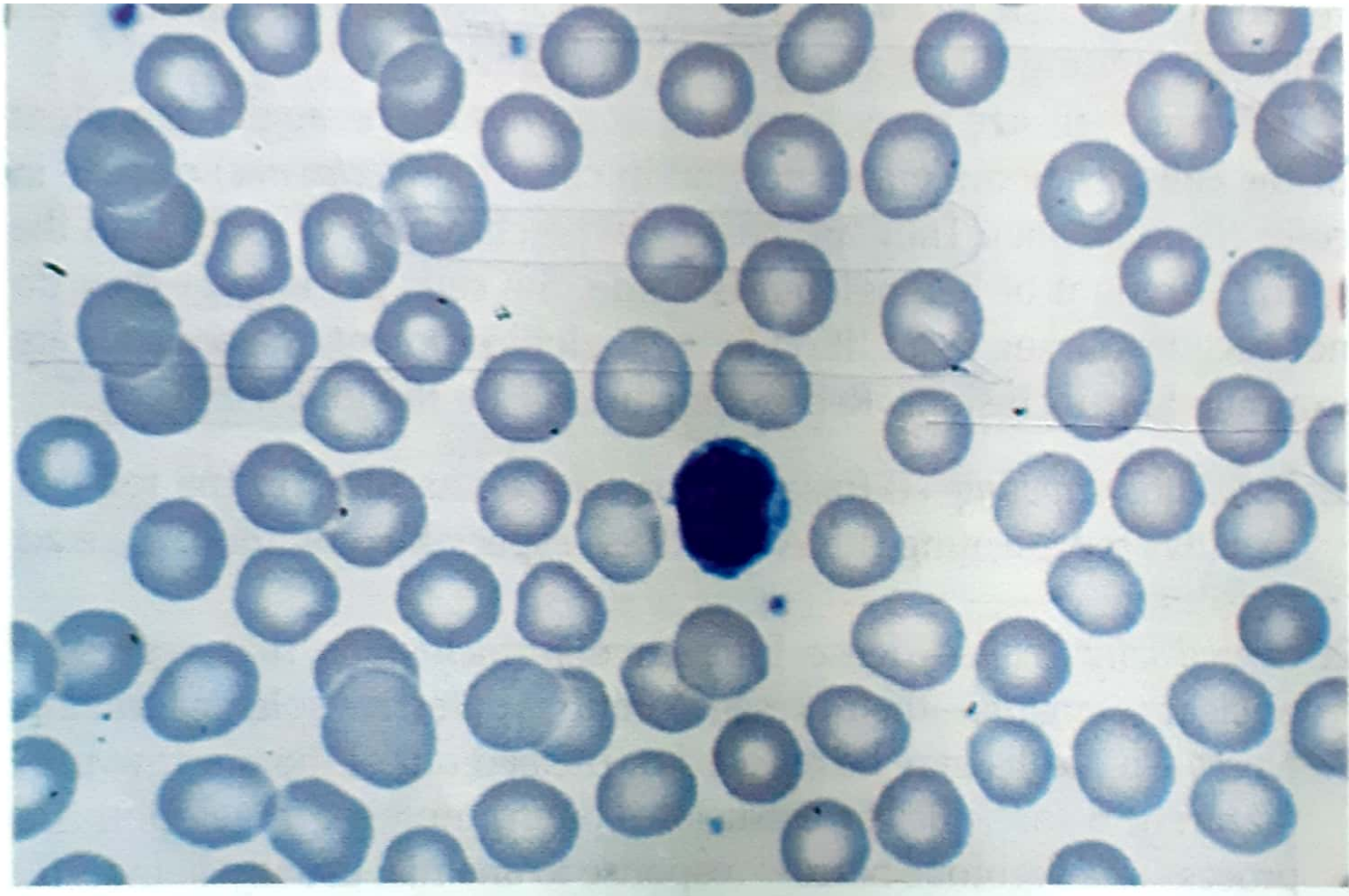


FIGURE 58-1 Normal peripheral blood smear. Small lymphocyte in center of field. Note that the diameter of the red blood cell is similar to the diameter of the small lymphocyte nucleus.

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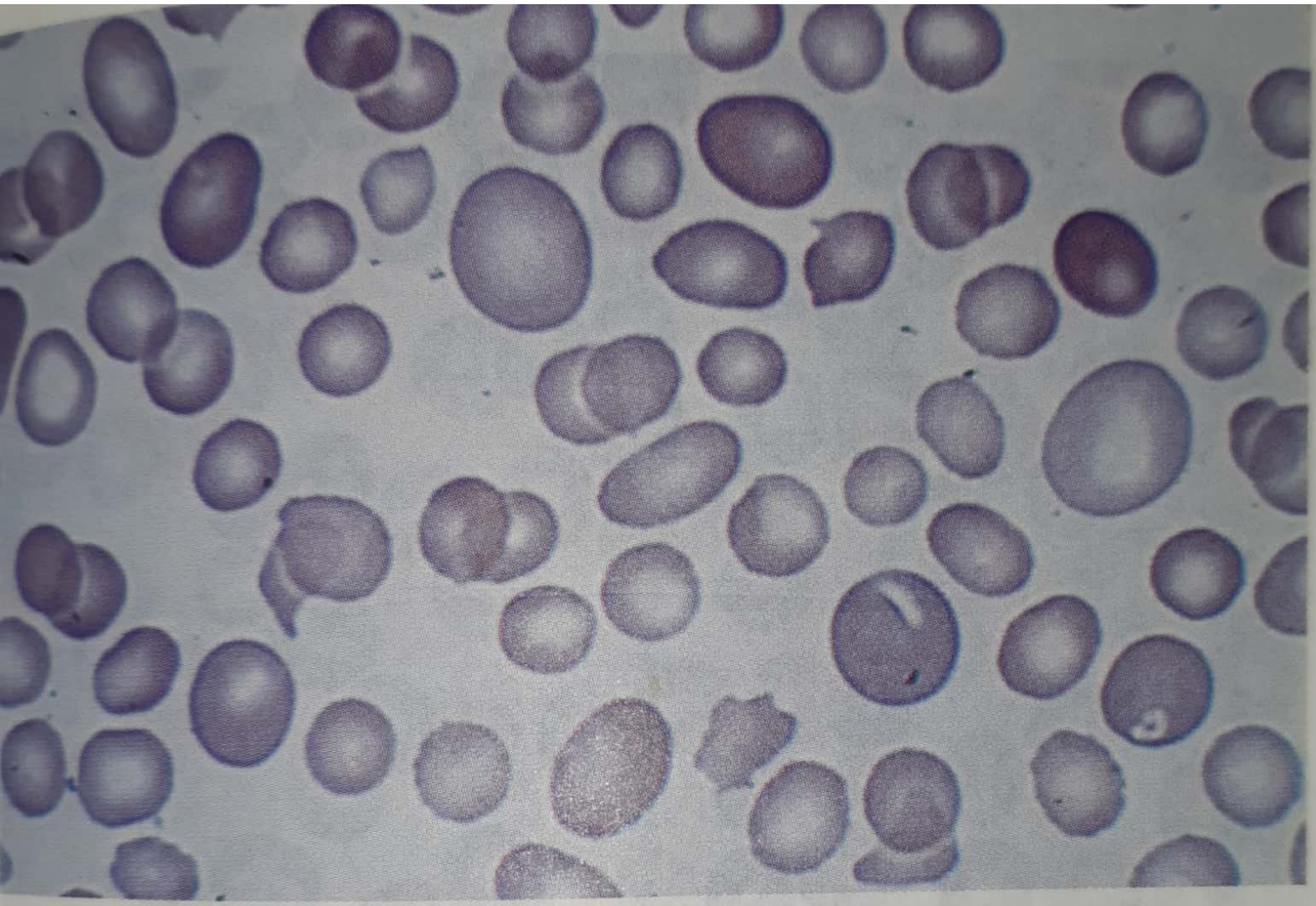


FIGURE 58-6 Macrocytosis. These cells are both larger than normal (mean corpuscular volume >100) and somewhat oval in shape. Some morphologists call these cells macroovalocytes.

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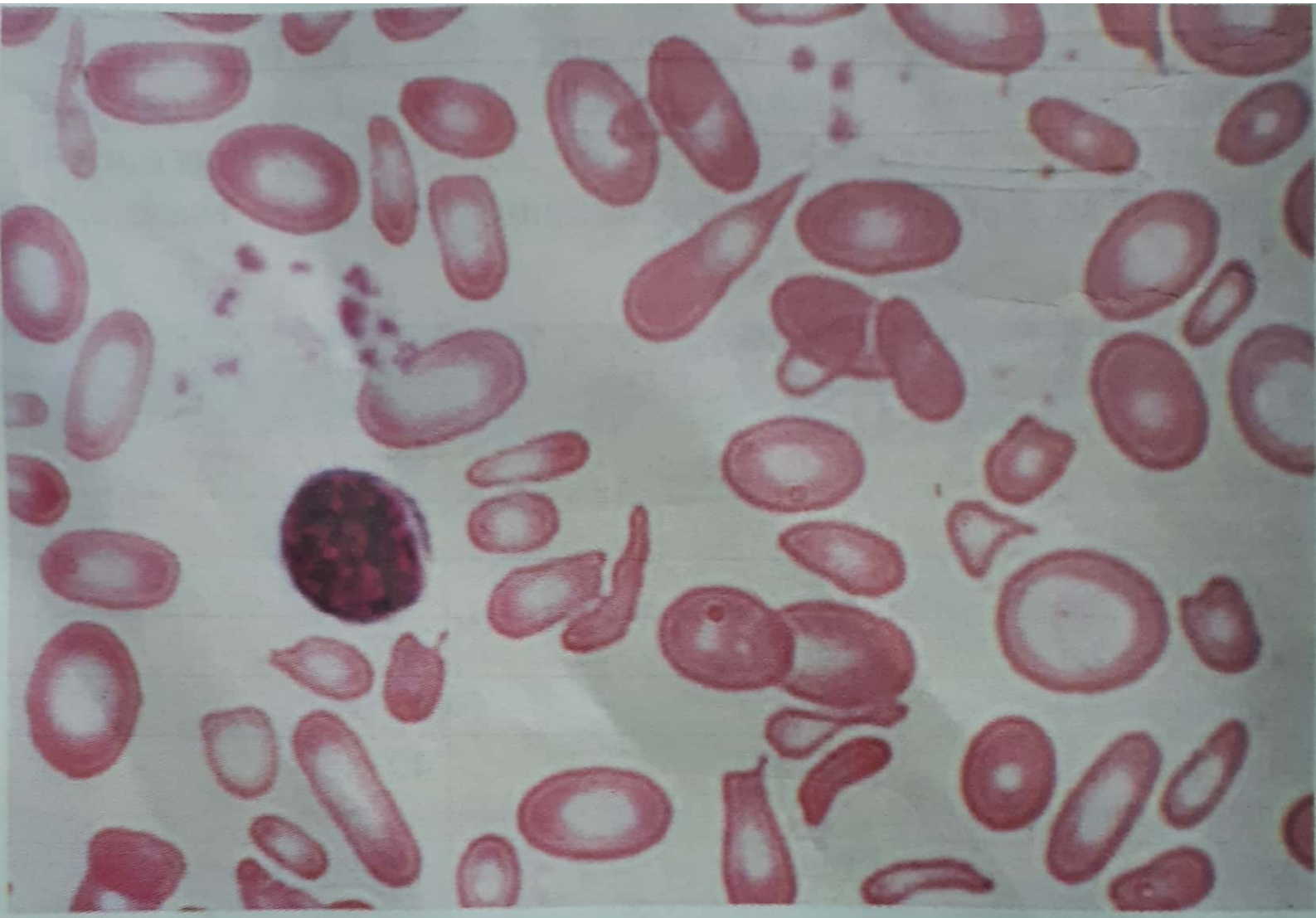
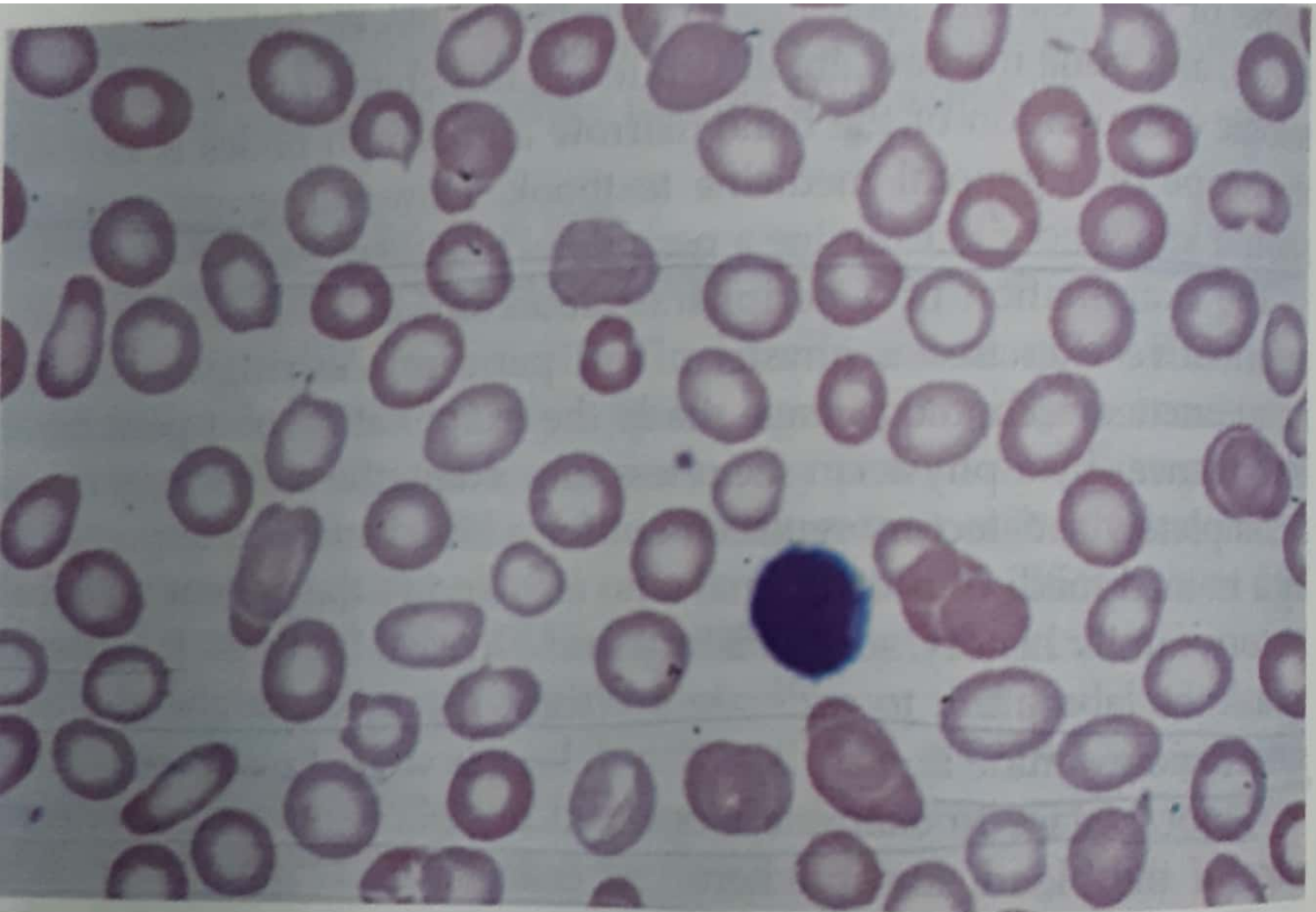


FIGURE 59-4 Severe iron-deficiency anemia. Microcytic and hypochromic red cells smaller than the nucleus of a lymphocyte associated with marked variation in size (anisocytosis) and shape (poikilocytosis). *(From RS Hillman et al: Hematology in Clinical Practice, 5th ed. New York, McGraw-Hill, 2010.)*

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IGURE 58-3 Hypochromic microcytic anemia of iron deficiency. Small ymphocyte in field helps assess the red blood cell size.

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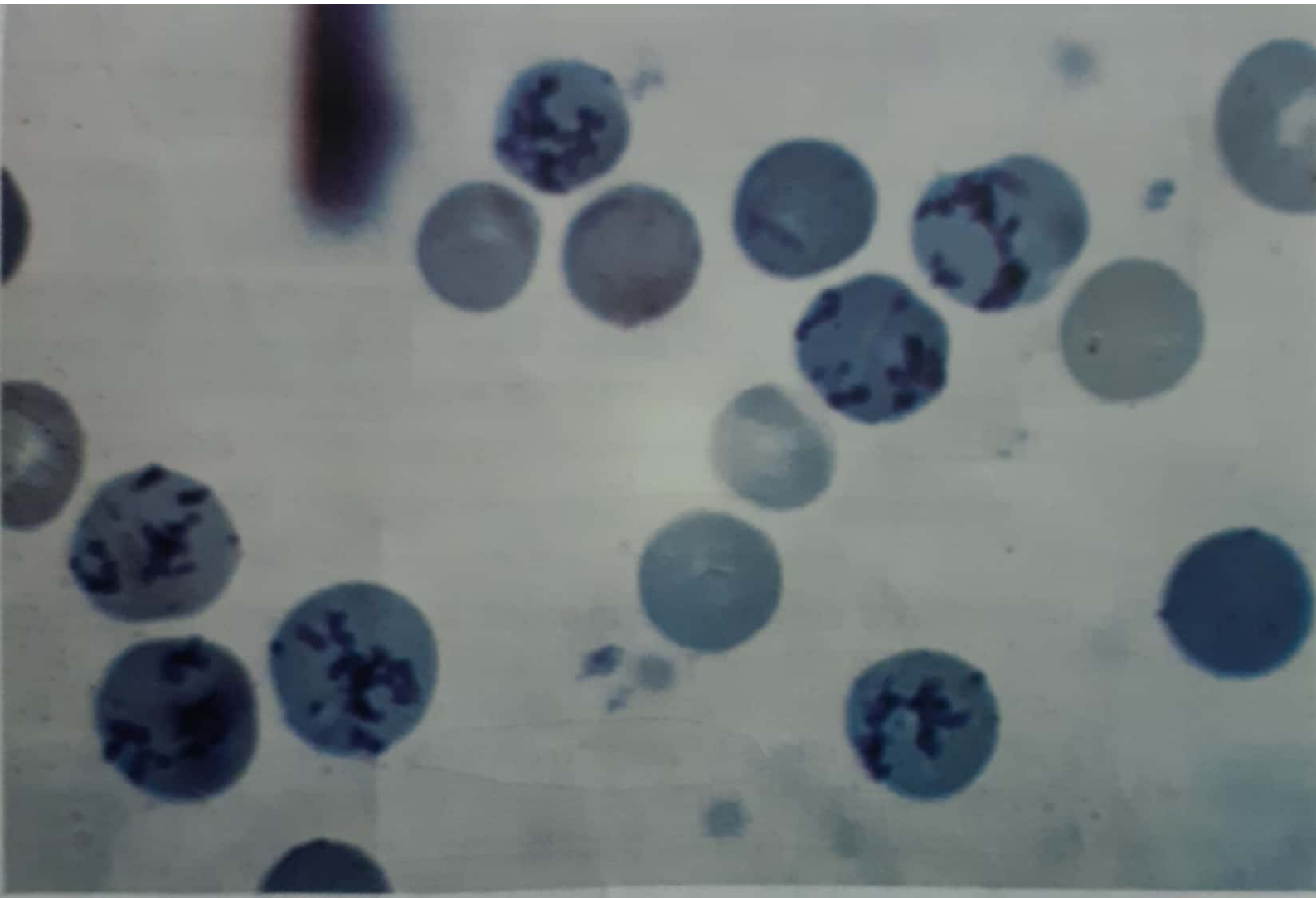
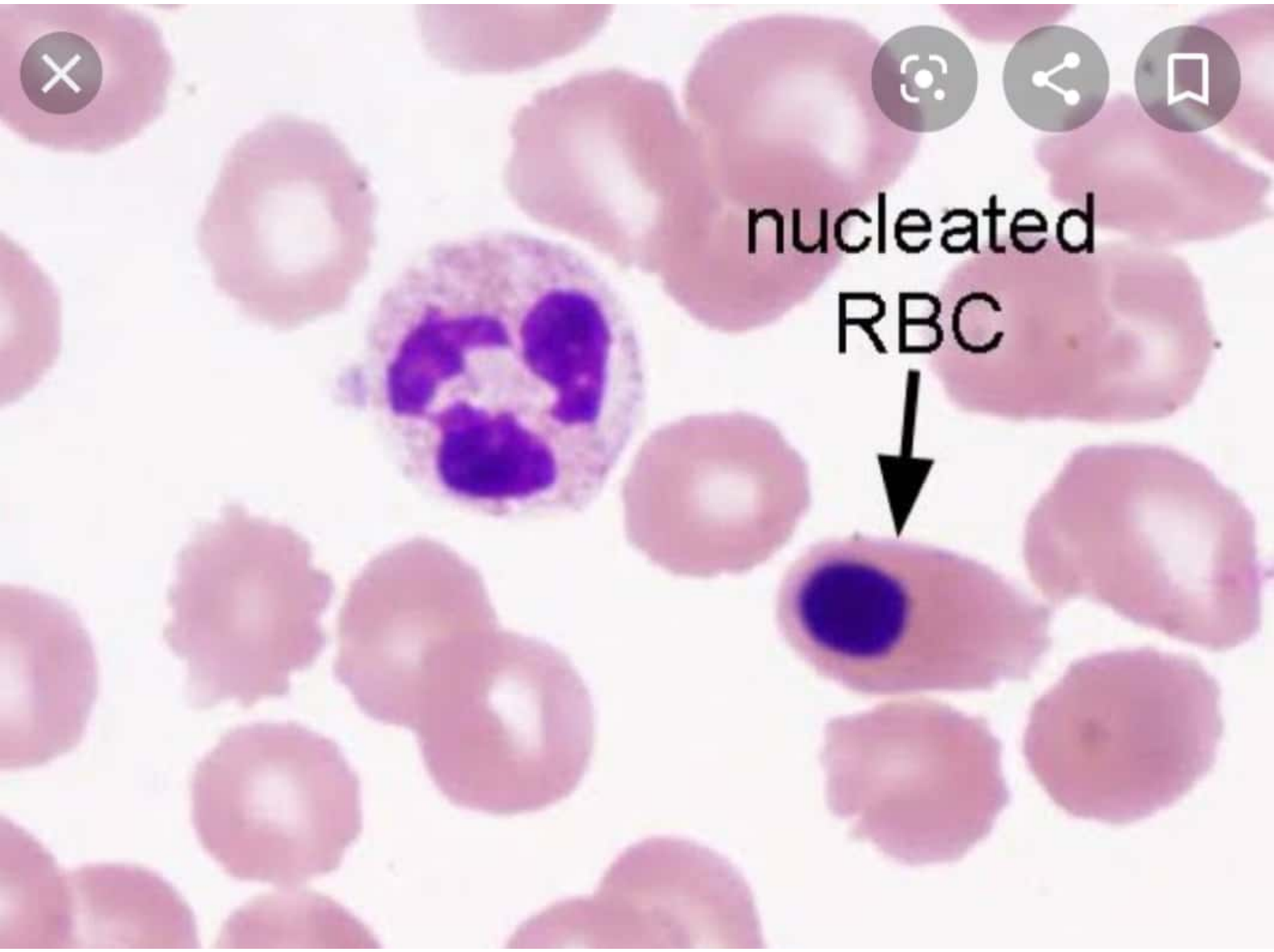


FIGURE 59-12 Reticulocytes. Methylene blue stain demonstrates residual RNA in newly made red cells. *(From RS Hillman et al: Hematology in Clinical Practice, 5th ed. New York, McGraw-Hill, 2010.)*

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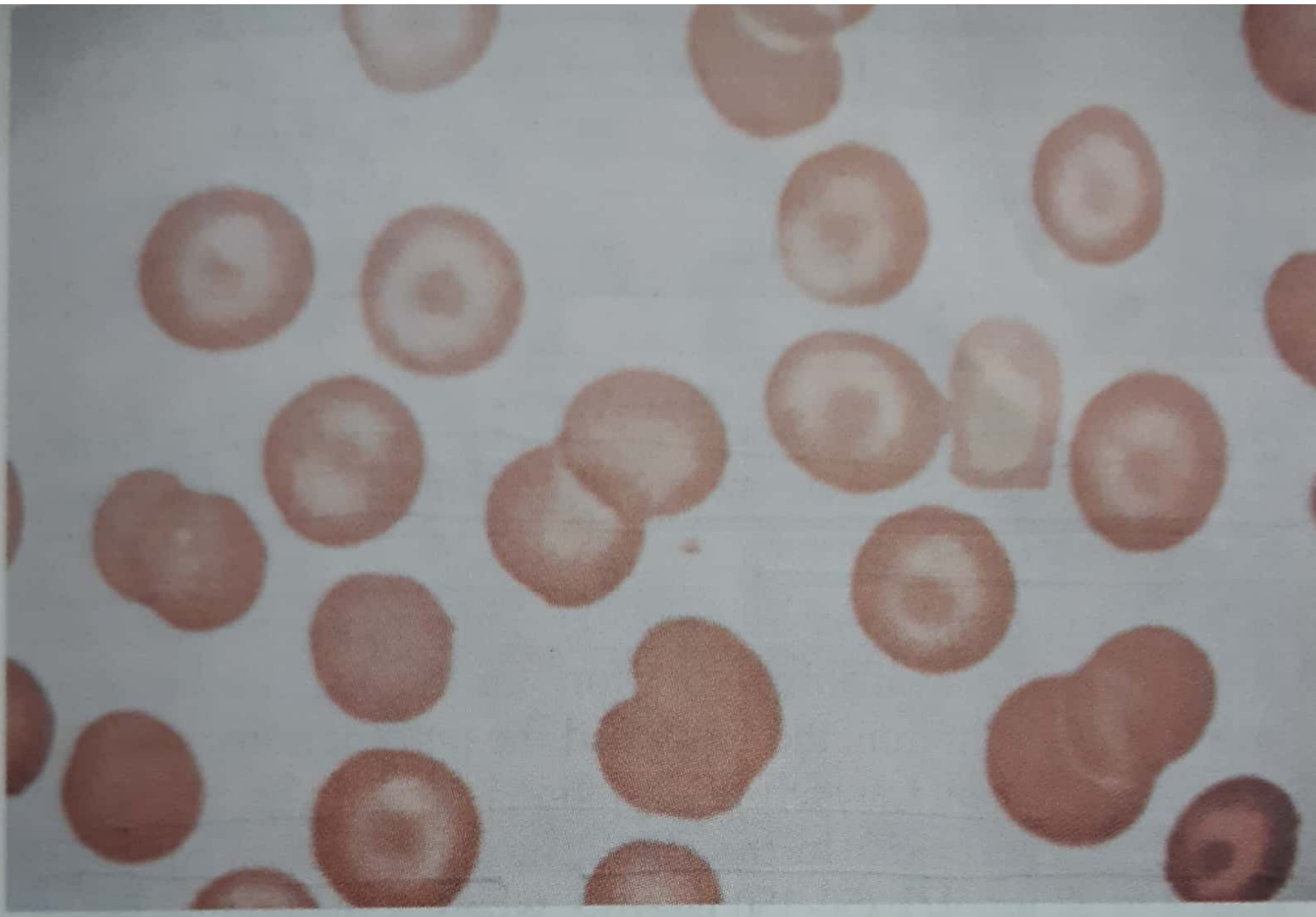


FIGURE 59-8 Target cells. Target cells have a bull's-eye appearance and are seen in thalassemia and in liver disease. *(From RS Hillman et al: Hematology in Clinical Practice, 5th ed. New York, McGraw-Hill, 2010.)*

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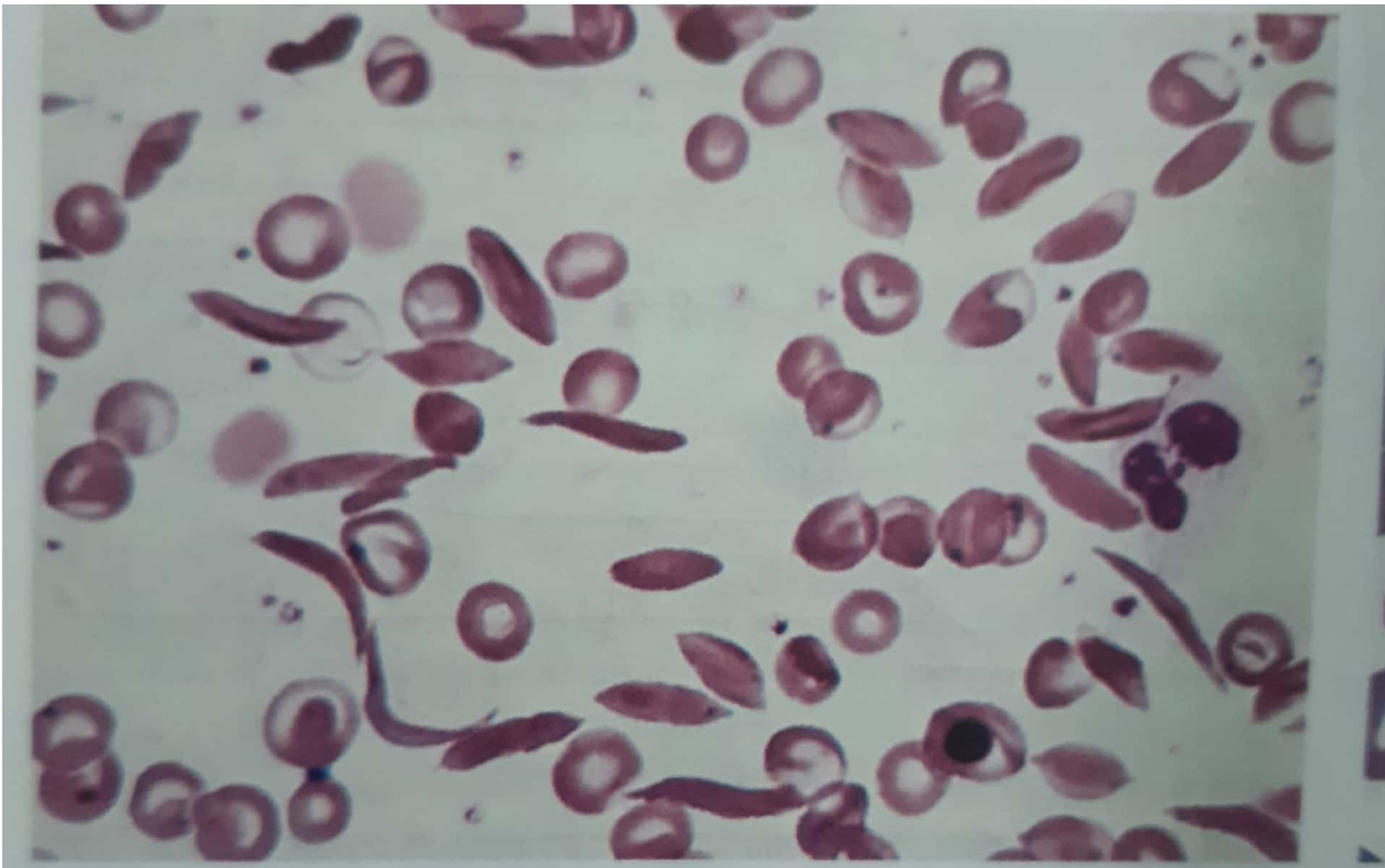
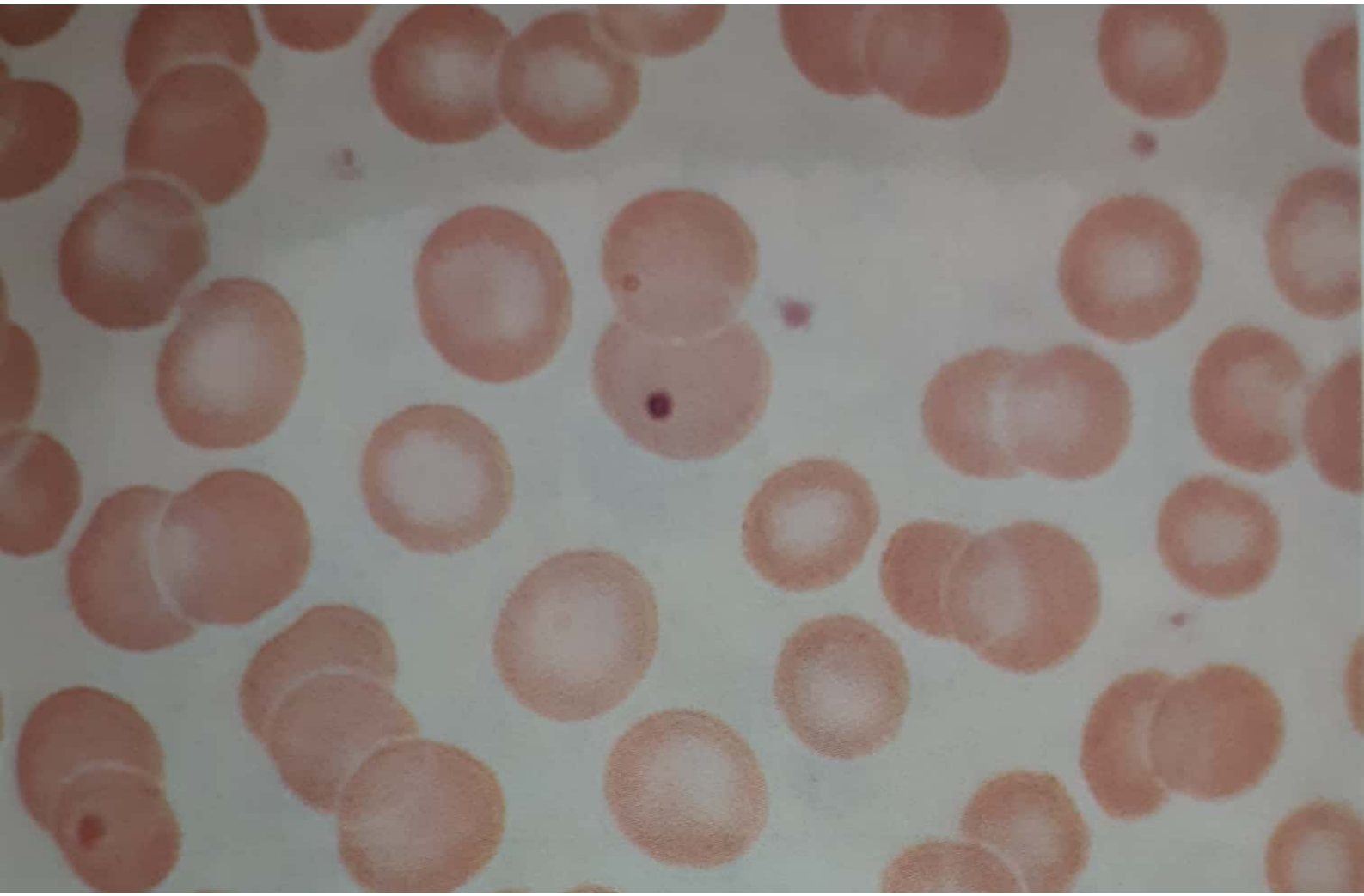


FIGURE 58-12 Sickle cells. Homozygous sickle cell disease. A nucleated red cell and neutrophil are also in the field.

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ANEMIA LAB INV.

- Establish Dx - (Hb, RBC count, Hct)
- Gain Basic Info - (Red cell Indices)
- Correlate (Platelet count, TWBC count, DLC)
- Way to Progression (PBS)

CHECKPOINT 1

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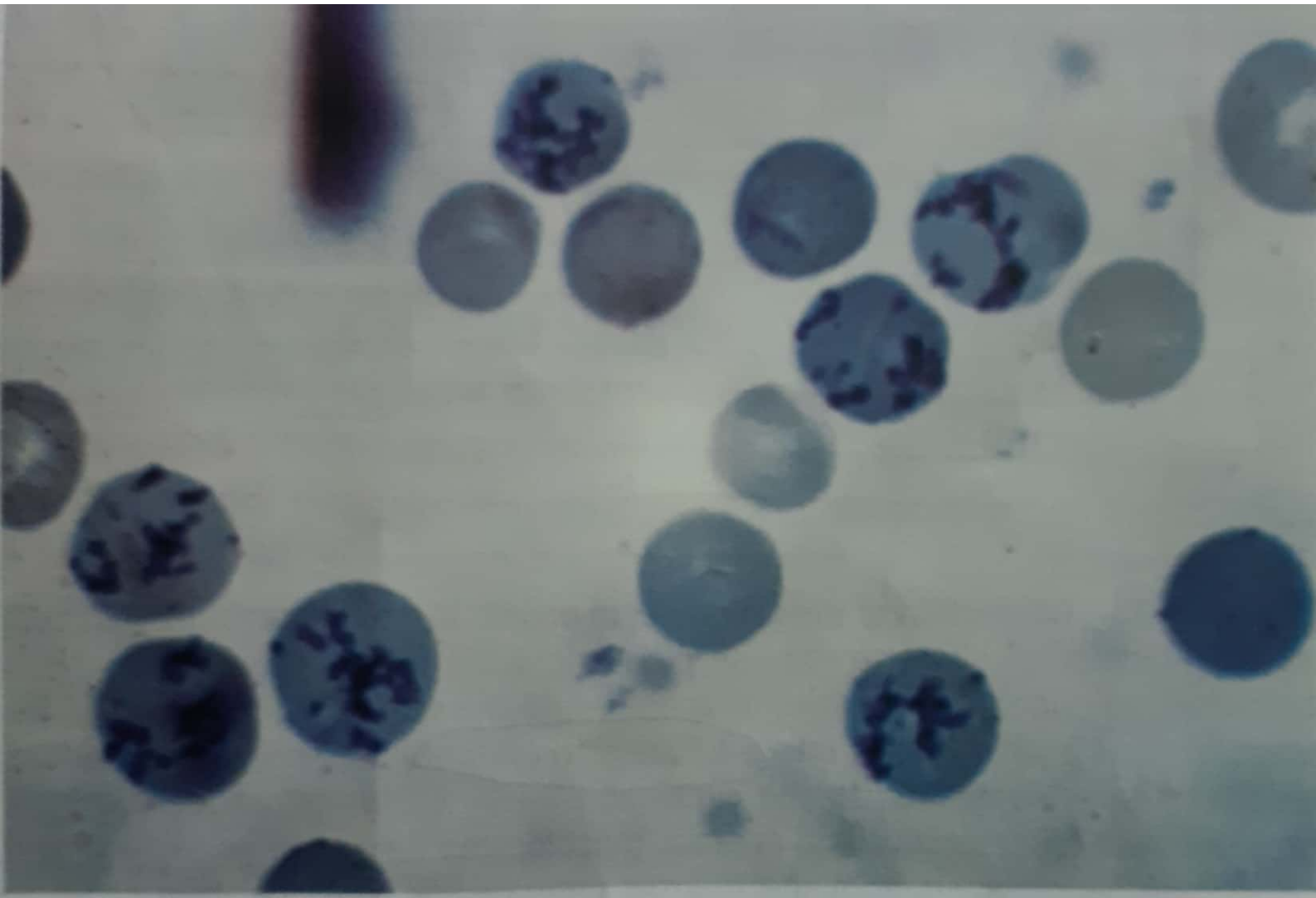


FIGURE 59-12 Reticulocytes. Methylene blue stain demonstrates residual RNA in newly made red cells. *(From RS Hillman et al: Hematology in Clinical Practice, 5th ed. New York, McGraw-Hill, 2010.)*

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TABLE 59-4 Calculation of Reticulocyte Production Index

Correction #1 for Anemia:

This correction produces the corrected reticulocyte count.

In a person whose reticulocyte count is 9%, hemoglobin 7.5 g/dL, and hematocrit 23%, the absolute reticulocyte count = 9 × (7.5/15) [or × (23/45)] = 4.5%

Note. This correction is not done if the reticulocyte count is reported in absolute numbers (e.g., 50,000/μL of blood)

Correction #2 for Longer Life of Prematurely Released Reticulocytes in the Blood:

This correction produces the reticulocyte production index.

In a person whose reticulocyte count is 9%, hemoglobin 7.5 gm/dL, and hematocrit 23%, the reticulocyte production index

$$= 9 \times \frac{(7.5/15)(\text{hemoglobin correction})}{2(\text{maturation time correction})} = 2.25$$

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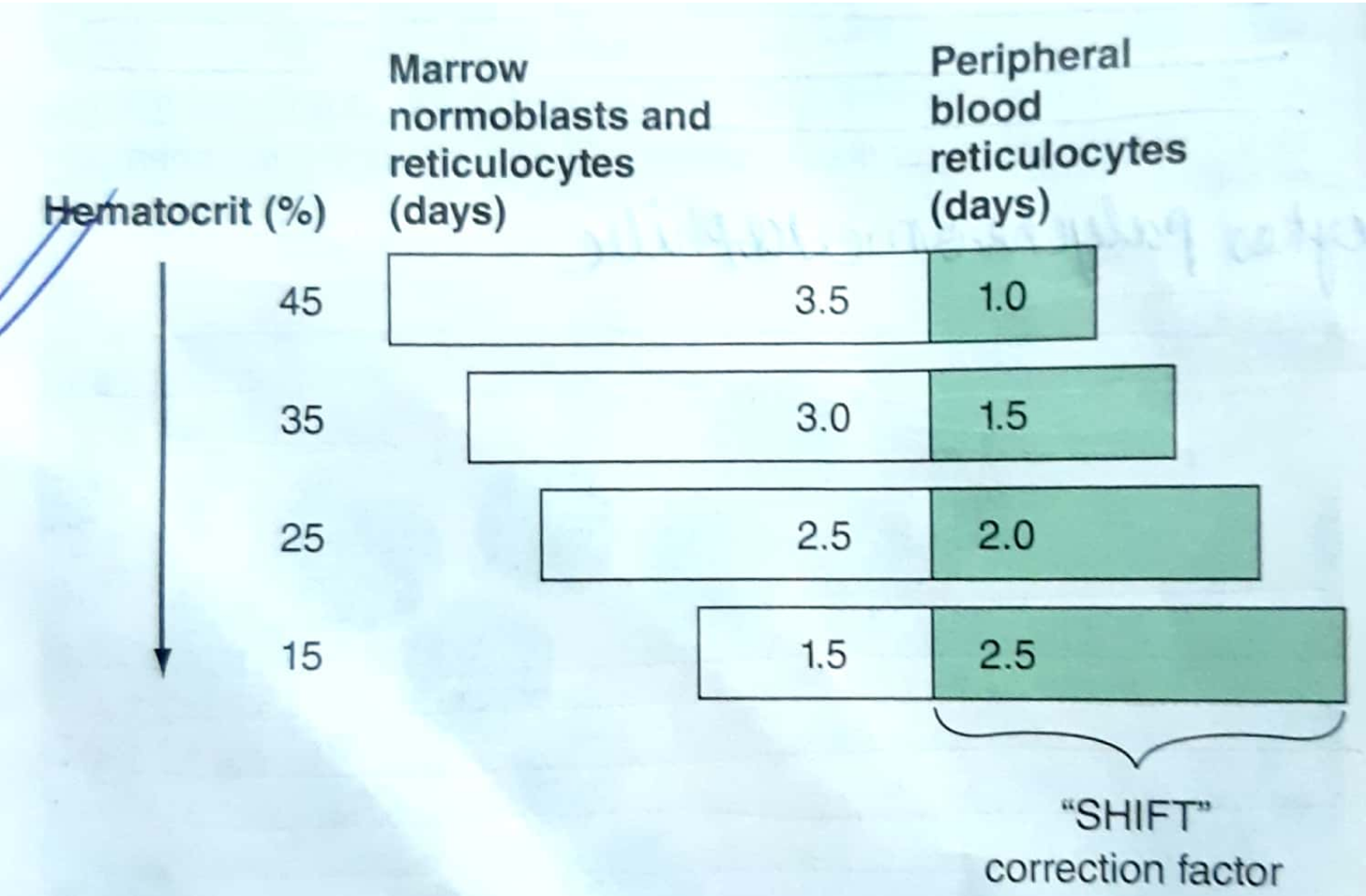


FIGURE 59-13 Correction of the reticulocyte count. To use the reticulocyte count as an indicator of effective red cell production, the reticulocyte number must be corrected based on the level of anemia and the circulating life span of the reticulocytes. Erythroid cells take ~4.5 days to mature. At a normal hemoglobin, reticulocytes are released to the circulation with ~1 day left as reticulocytes. However, with different levels of anemia, reticulocytes (and even earlier erythroid cells) may be released from the marrow prematurely. Most patients come to clinical attention with hematocrits in the mid-20s, and thus a correction factor of 2 is commonly used because the observed reticulocytes will live for 2 days in the circulation before losing their RNA.

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TABLE 59-5 Normal Marrow Response to Anemia

HEMOGLOBIN	PRODUCTION INDEX	RETICULOCYTE COUNT
15 g/dL	1	50,000/μL
11 g/dL	2.0–2.5	100–150,000/μL
8 g/dL	3.0–4.0	300–400,000/μL

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ANEMIA LAB INV.

- Establish Dx - (Hb, RBC count, Hct)
- ↓
- Gain Basic Info - (Red cell Indices)
- ↓
- Correlate (Platelet count, TWBC count, DLC)
- ↓
- Way to Go forward (PBS)

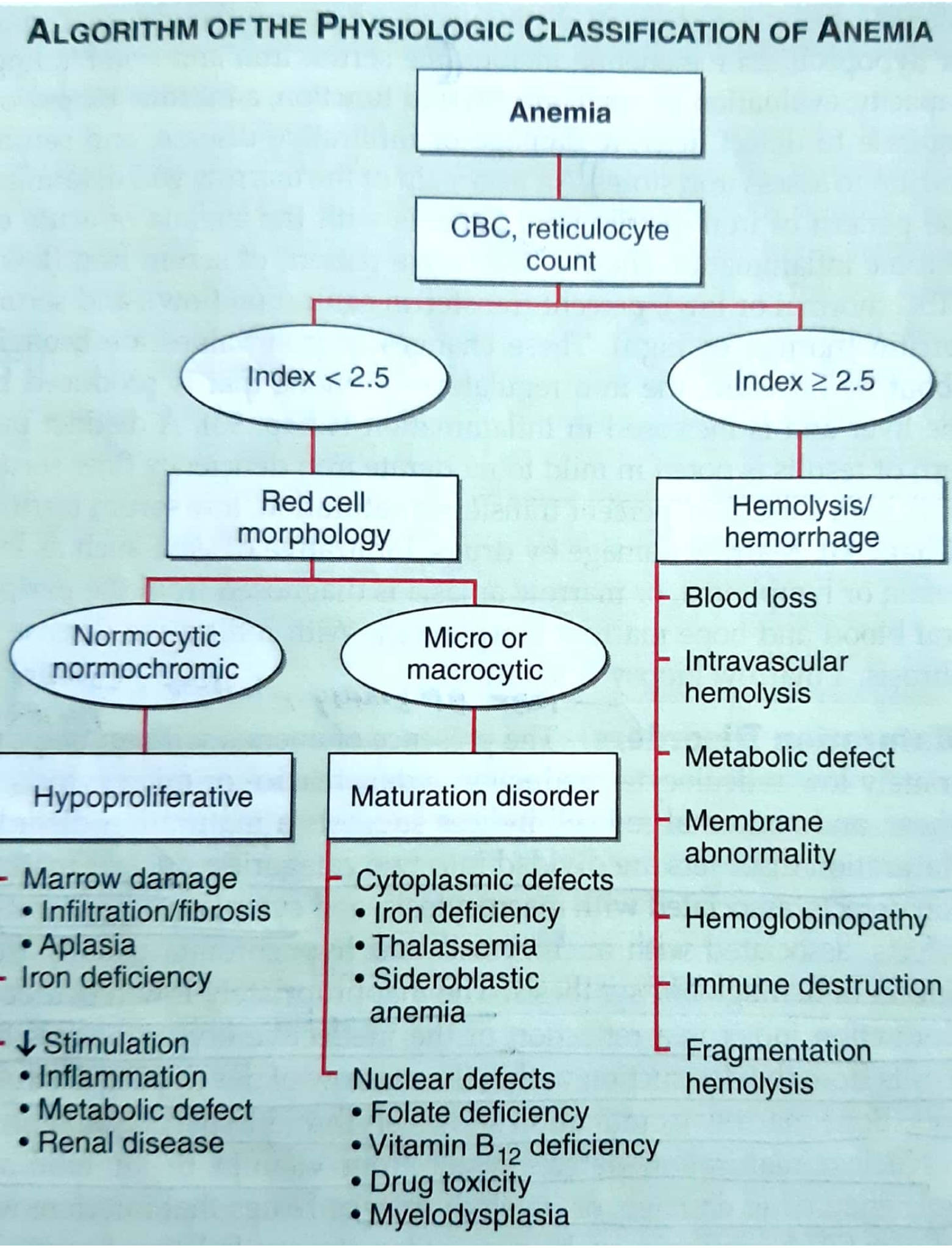
CHECKPOINT 1

Reticulocyte count

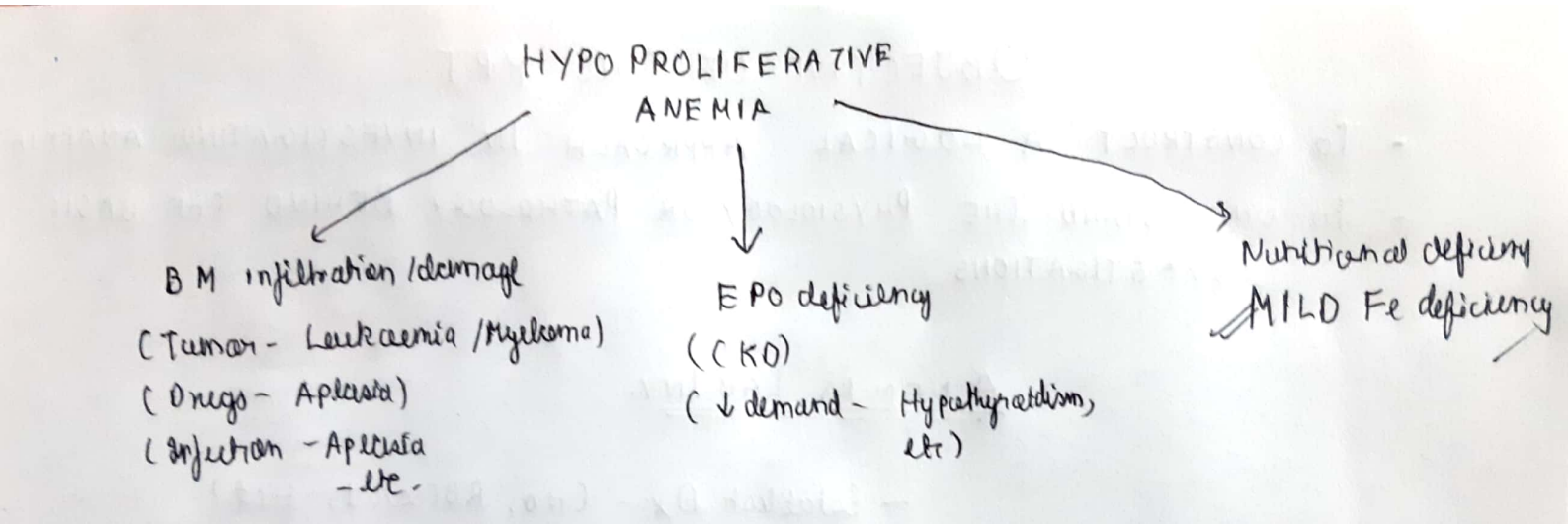
CHECKPOINT 2

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Algorithm of the Physiologic Classification of Anemia



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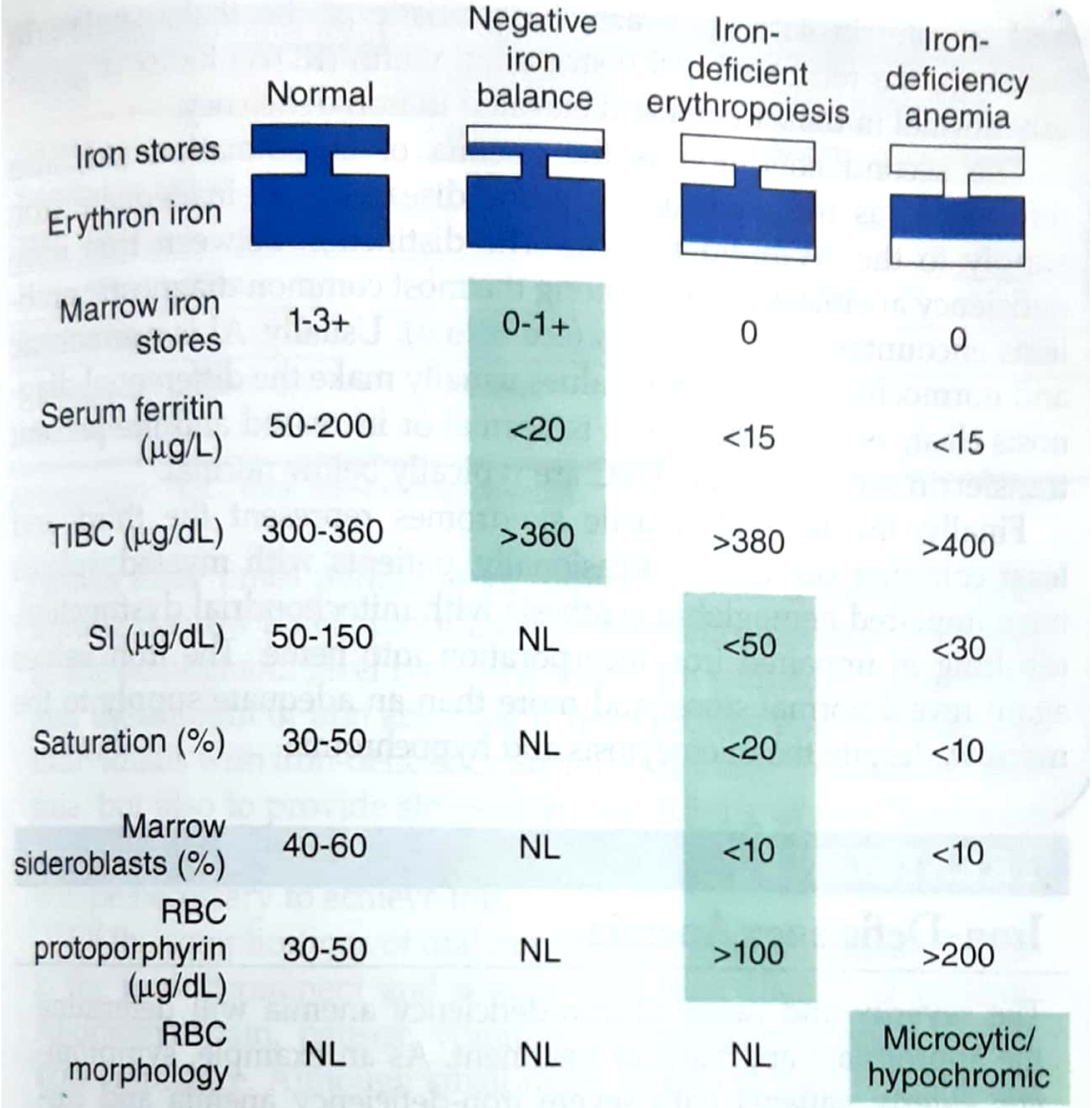


FIGURE 93-2 Laboratory studies in the evolution of iron deficiency. Measurements of marrow iron stores, serum ferritin, and total iron-binding capacity (TIBC) are sensitive to early iron-store depletion. Iron-deficient erythropoiesis is recognized from additional abnormalities in the serum iron (SI), percent transferrin saturation, the pattern of marrow sideroblasts, and the red blood cell (RBC) protoporphyrin level. Patients with iron-deficiency anemia demonstrate all the same abnormalities plus hypochromic microcytic anemia. *(From RS Hillman, CA Finch: The Red Cell Manual, 7th ed. Philadelphia, F.A. Davis and Co, 1996, with permission.)*

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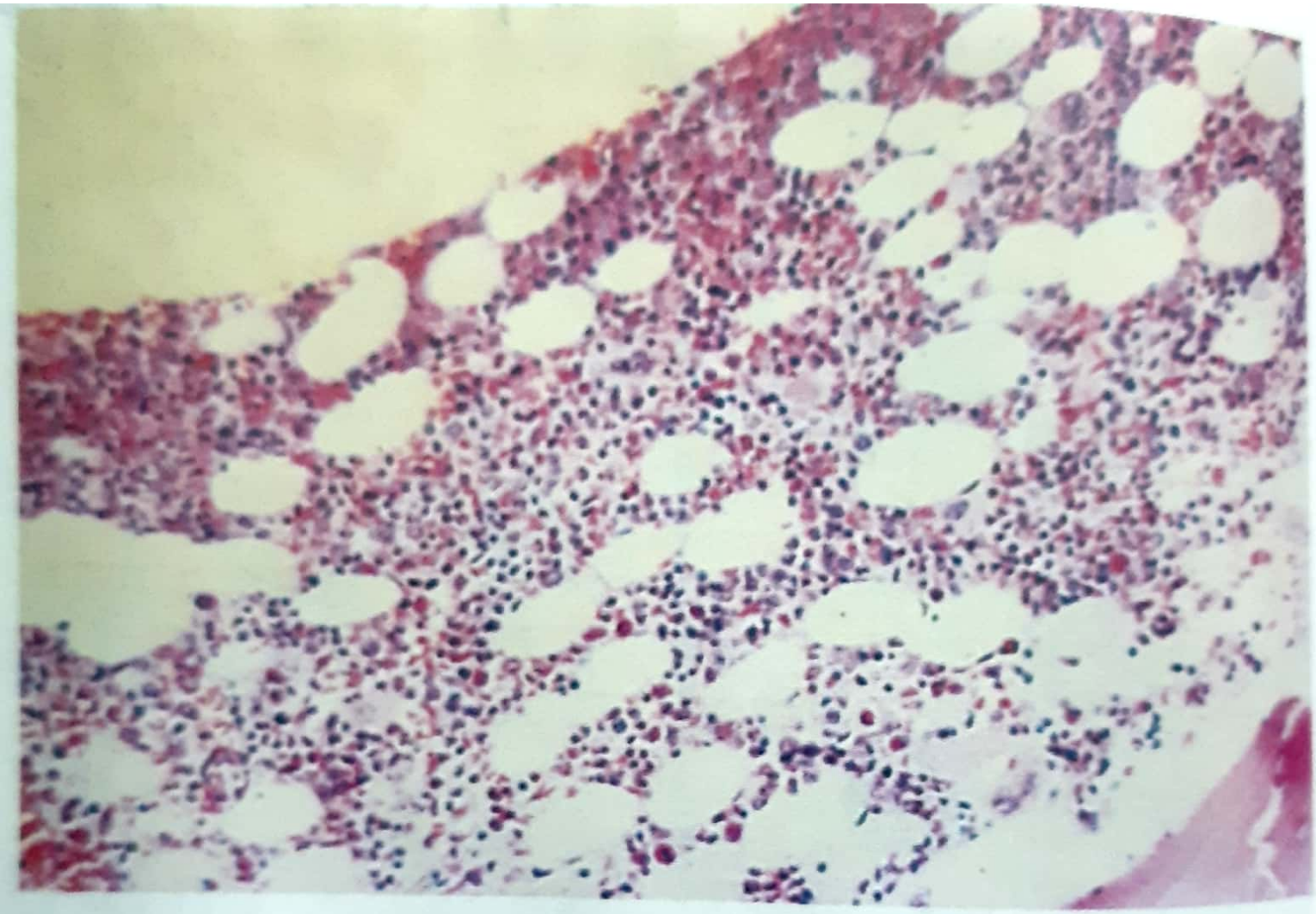


FIGURE 59-14 Normal bone marrow. This is a low-power view of a section of a normal bone marrow biopsy stained with hematoxylin and eosin (H&E). Note that the nucleated cellular elements account for ~40–50% and the fat (clear areas) accounts for ~50–60% of the area. *(From RS Hillman et al: Hematology in Clinical Practice, 5th ed. New York, McGraw-Hill, 2010.)*

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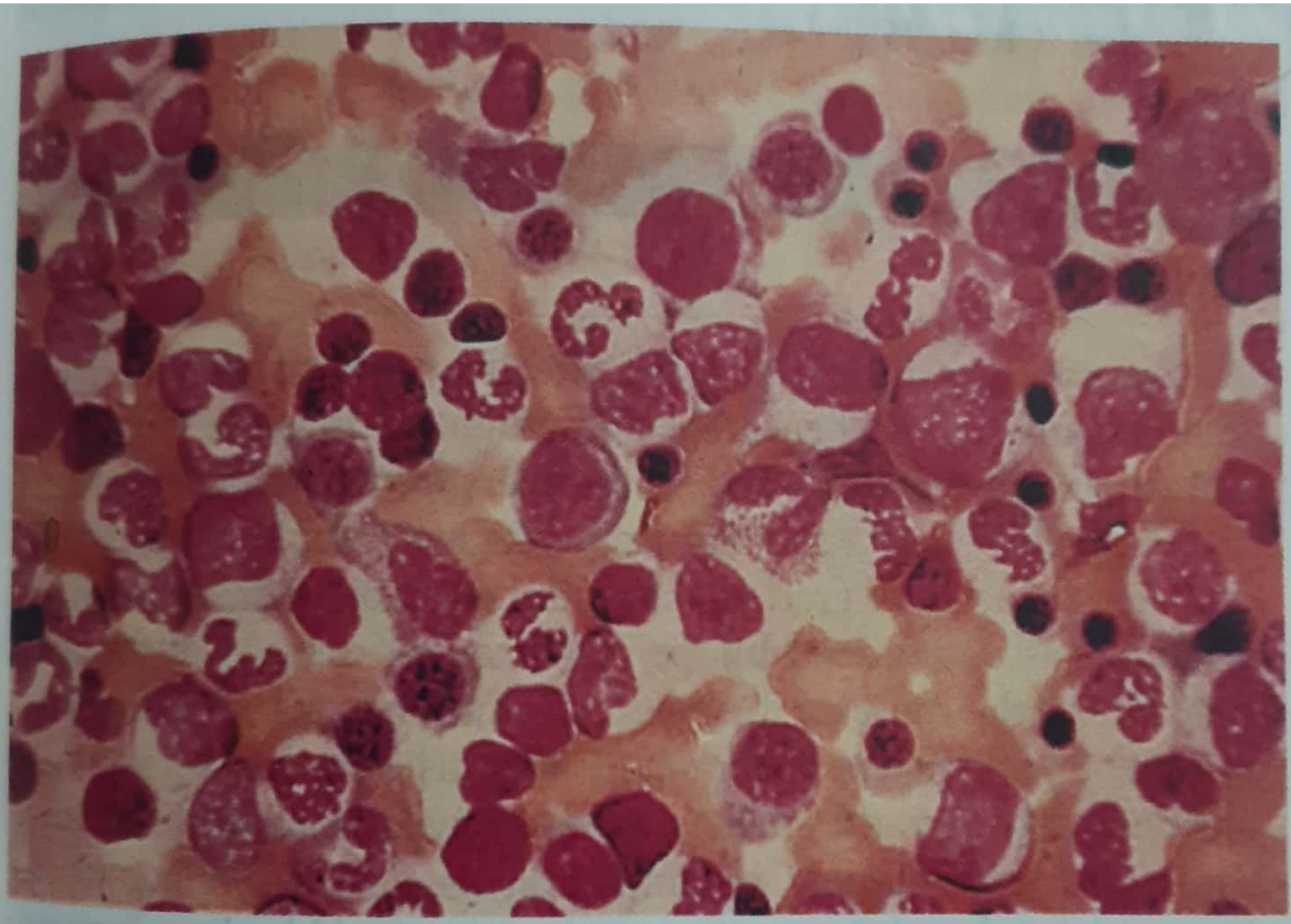
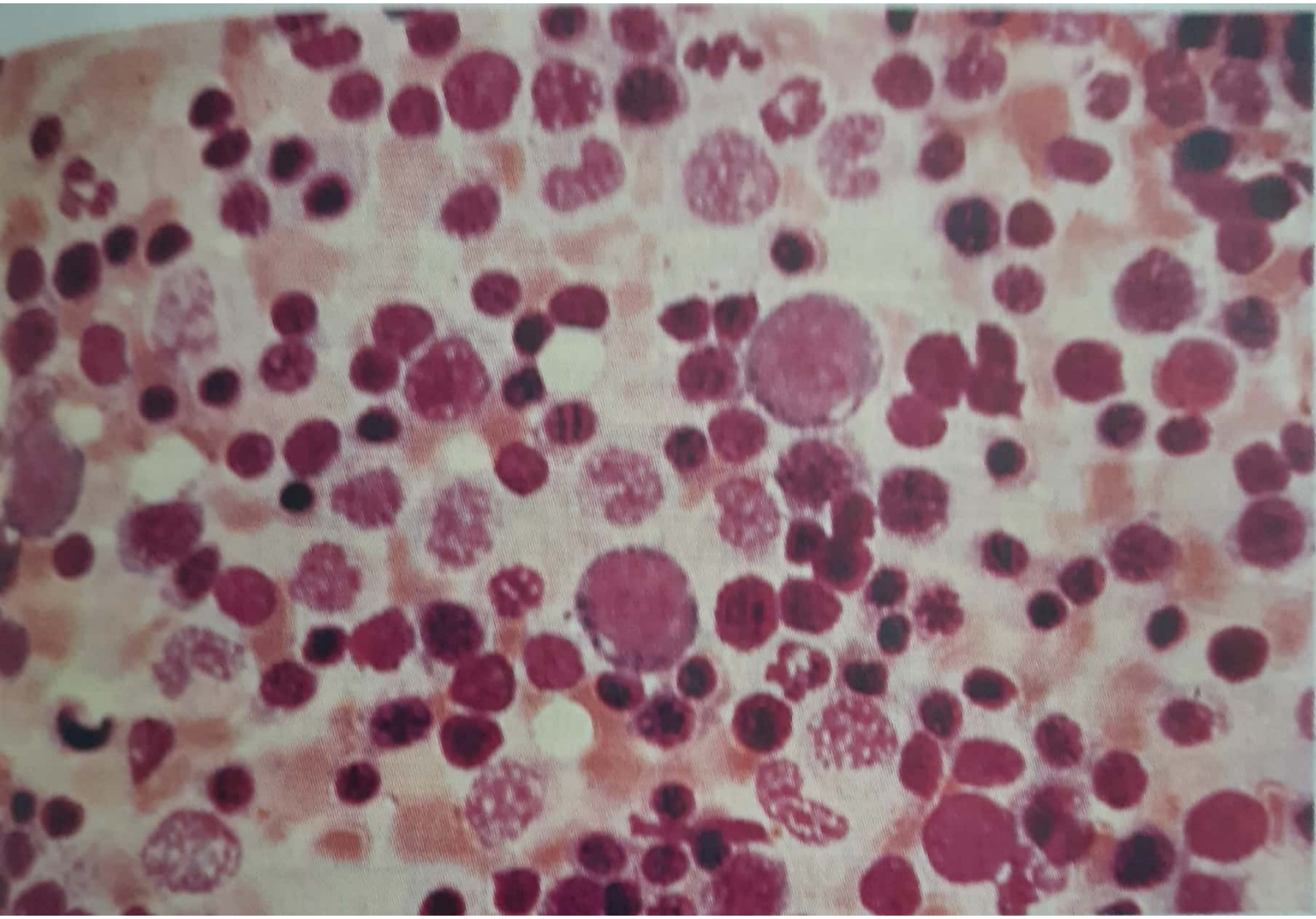
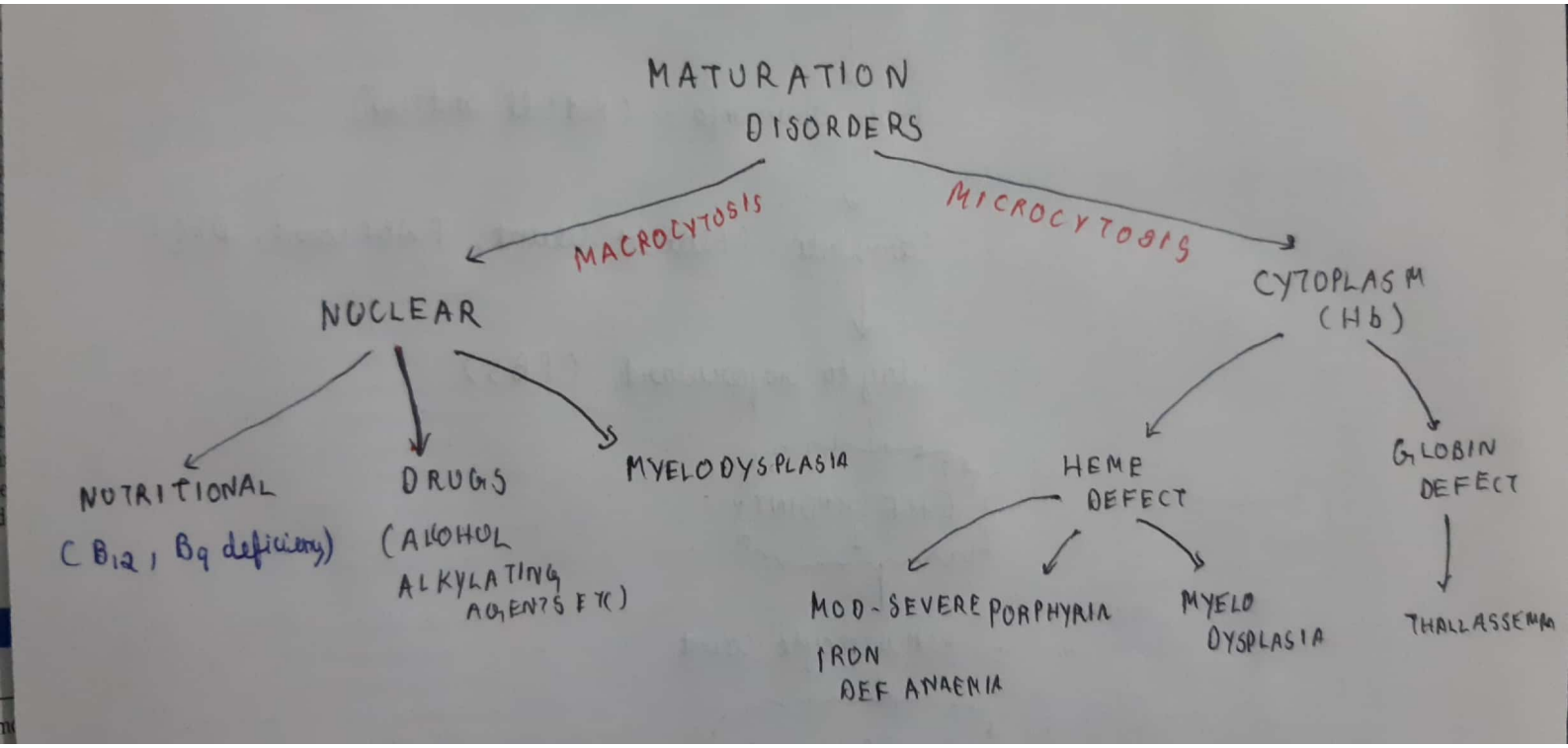


FIGURE 59-16 Myeloid hyperplasia. This marrow shows an increase in the fraction of cells in the myeloid or granulocytic lineage as might be seen in a normal marrow responding to infection. The myeloid/erythroid (M/E) ratio is >3:1. *(From RS Hillman et al: Hematology in Clinical Practice, 5th ed. New York, McGraw-Hill, 2010.)*

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TABLE 96-2 Features Common to Most Patients with a Hemolytic Disorder

General examination	Jaundice, pallor
Other physical findings	Spleen may be enlarged; bossing of skull in severe congenital cases
Hemoglobin level	From normal to severely reduced
MCV, MCH	Usually increased
Reticulocytes	Usually Increased
Bilirubin	Almost always increased (mostly unconjugated)
LDH	Increased (up to 10× normal with intravascular hemolysis)
Haptoglobin	Reduced to absent if hemolysis is at least in part intravascular)

Abbreviations: LDH, lactate dehydrogenase; MCH, mean corpuscular hemoglobin; MCV, mean corpuscular volume.

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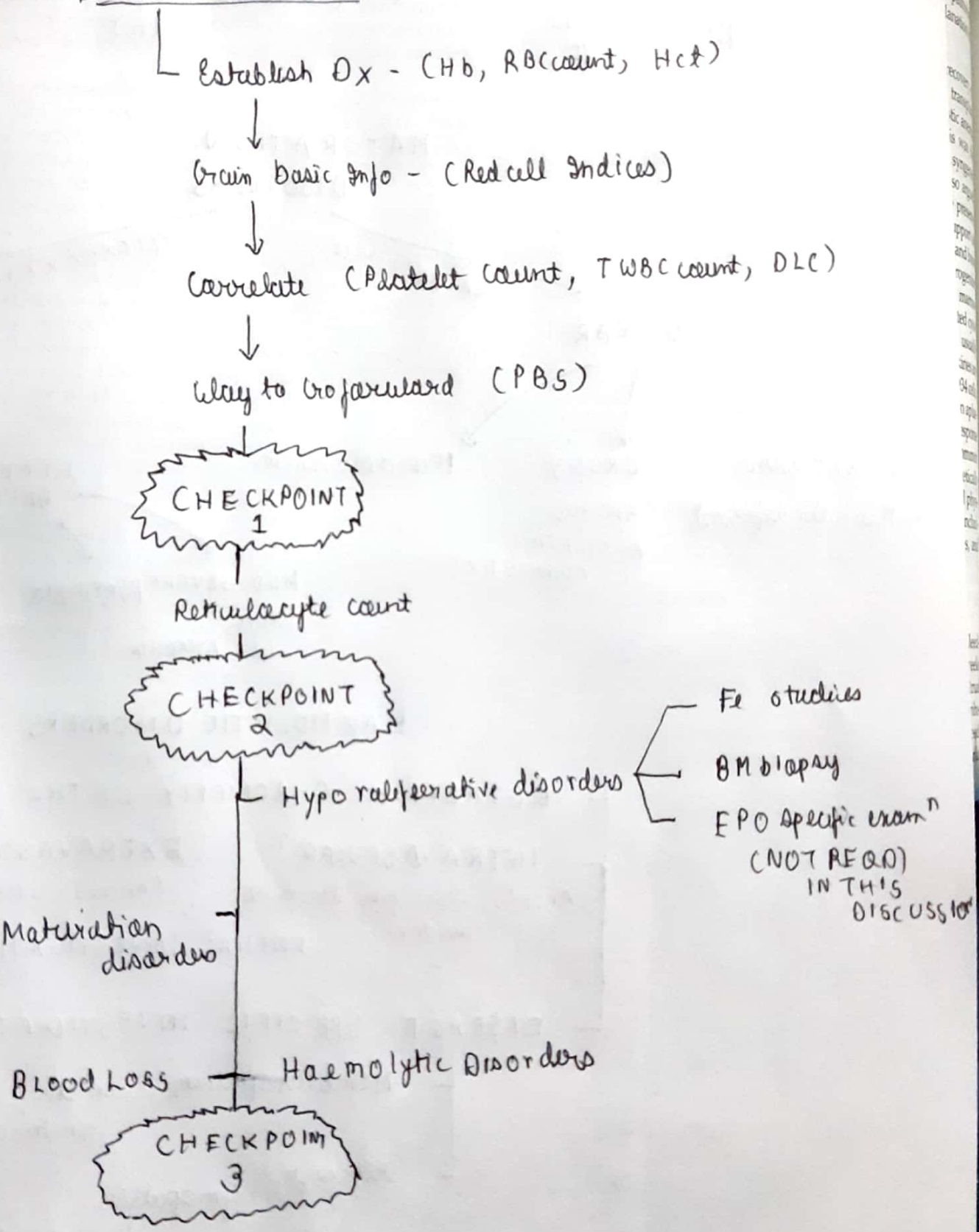
HAEMOLYTIC DISORDERS

- ESTABLISH CHRONICITY OF THE DISEASE (IF PRESENT)
- INTRAVASCULAR? EXTRAVASCULAR?
  (Anemia, Urine-Hb, Blood-Hb, Jaundice) (Anemia, Jaundice, Splenomegaly)

  INITIAL INVESTIGATIONS
- DISEASE SPECIFIC INVESTIGATION
  - Membranopathies (osmotic fragility test, Gel Electrophoresis, Genetic analysis etc)
  - ~~Glycolytic~~ Enzymopathies
    - Glycolytic (Enzyme assays, Genetic analysis)
    - Redox
  - Hb pathies - (Gel Electrophoresis, etc)
  - ACQUIRED? - (Flow cytometry, Genetic analysis etc?)

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ANEMIA LAB INV.



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Q. WHO cut off for Anemia in ♀?

Q. Haematocrit value for ♂?

Q. For Mod anemia the ____ times ↑ in Red Cell Production due to EPO.

Q. Production Index cut off used in diagnostic algorith?

Q. The 3 basic pathologies in Hypoproliferative anemia?

Q. Nuclear maturation disorder causes MCV ____?

Q. TIBC ____ on Iron deficiency anemia

Q. 2 storage forms of Iron and their tests?

Q. Iron dificiency causes both ____ and ____ type anemia.

Q. In Extravascular haemolysis the clinical triad is?

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1. 12 g/dl
2. 47 ± 7%
3. 2 - 3 Hours
4. 2.5
5. Hypoproliferative:
   BM damage
   Mild Fe deficiency
   ↓ EPO sensitivity
6. ↑
7. ↑
8. Ferritin - Serum Ferritin
   Haemosidrin - BM Siderablasts & Haemosiderin
9. Hypoproliferative and Maturation defect
10. Anemia Jaundice Splenomegaly

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